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Short communication

Adverse psychosocial work exposures and 2-year incident prescription of psychotropic medications: a prospective French study using the national healthcare reimbursement database

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ABSTRACT

Objective: Despite the abundant literature on the associations between psychosocial work exposures and mental health outcomes, prospective studies using medico-administrative data are lacking. The objective was to study the associations between various psychosocial work exposures and incident prescription of psychotropic medications.

Methods: The study relied on three waves (2010, 2012, 2014) of the French ESPS survey (Health, Health Care and Insurance Survey) (baseline data), linked to the national database of healthcare reimbursements (follow-up data). Each survey wave was conducted among a randomly selected national sample of people aged 18+. The study included a total of 8689 observations among 7695 workers without psychotropic medication prescription within the 6 months preceding survey wave. Ten psychosocial work exposures (quantitative demands, tensions with the public, freedom, possibilities for learning new things, colleague support, recognition, salary satisfaction, job insecurity, temporary contract, and redundancy plan) were assessed at each survey wave. Covariates included gender, age, marital status, employee/self-employed worker, private/public sector, occupation, and full/part-time work. The outcome was psychotropic medication prescription within the 2 years following survey wave. Mixed effects Cox regression modelling was used.

Results: Seven of the ten psychosocial work exposures were associated with incident psychotropic medication prescription. Associations were observed more with incident prescription of antidepressants and anxiolytics than with hypnotics/sedatives. The number of psychosocial work exposures increased the risk linearly, except for hypnotics/sedatives. No gender interaction was found for any of these associations.

Conclusion: The findings may support that poor psychosocial working conditions increase the risk of subsequent psychotropic medication prescription.

1. Introduction

There is a growing literature showing that psychosocial work exposures are associated with health problems, especially with mental health problems and disorders (Niedhammer et al., 2021; Rugulies et al., 2023). Literature reviews including prospective studies show consistent evidence for the associations between factors related to the two work stress models of job strain and effort-reward imbalance, but also the factors of organisational injustice, job insecurity, and bullying/violence at work with various measures of mental disorders, especially depressive disorders and antidepressant treatment (Niedhammer et al., 2021; Rugulies et al., 2023). Psychosocial work exposures could be associated with mental disorders other than depression, such as anxiety and sleep disorders, underlining the interest in exploring all types of psychotropic medications, including anxiolytics and hypnotics/sedatives which have been much less studied than antidepressants (Blomqvist et al., 2020; Conway et al., 2025; Milner et al., 2019).

This prospective study aimed to examine the associations of various psychosocial work exposures, and multiple exposures, with incident prescription of psychotropic medications and their different therapeutic classes using data from a large French survey linked to the national healthcare reimbursement database.

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2. Methods

2.1. Study sample

The study was based on three waves (2010, 2012, 2014) of the French ESPS survey (Enquête Santé et Protection Sociale - Health, Health Care and Insurance Survey) used as baseline data. Each survey wave was conducted among a national sample of people aged 18+, randomly selected among the beneficiaries of the French national health insurance system. A subsample was interviewed twice (2010 and 2014). For the objectives of the study, we included in the study sample the ESPS survey participants aged 18–65, who were working, who responded to the questionnaire on working conditions (>70 %), with follow-up through the national database of healthcare reimbursements, i.e. SNDS which is the National Health Data System (>83 %), and without psychotropic medication prescription within the 6 months preceding survey wave. The flow diagram of the study sample selection resulted in a total of 8689 observations among 7695 workers (Fig. S1). The present study was approved by French ethics committees.

2.2. Measurements

The national database of healthcare reimbursements provides data on all pharmacy-dispensed medications with information on their prescription date and the prescribed medications coded using the Anatomical Therapeutic Chemical (ATC) classification system. We focused on antidepressants (ATC code: N06A), anxiolytics (ATC code: N05B), and hypnotics/sedatives (ATC code: N05C). Participants who had a prescription for any of these psychotropic medications within the 6 months preceding survey wave were excluded. Among the others, those who had a prescription of any psychotropic medications within the two years following survey wave were classified as incident cases.

The questionnaire on working conditions was the same at each survey wave and included the assessment of 10 psychosocial work exposures. Workers were asked about their employment contract (temporary/permanent) and their exposure to redundancy plan within the last 12 months (yes/no). Redundancy plan in France can be defined by collective redundancy within the framework of economic difficulties of companies. A 4-point Likert scale was used to collect the exposure frequency (always, often, sometimes, never) to each of the other psychosocial work exposures, and workers were then classified into 'low' (sometimes/never) or 'high' (always/often) frequency categories. For quantitative demands, tensions with the public and job insecurity, workers with 'high' frequency were defined as exposed, while 'low' frequency was the exposed group for freedom at work, possibilities for learning new things, support from colleagues, recognition at work, and satisfaction with salary. The number of psychosocial work exposures was calculated by counting the factors for which each individual was classified as exposed, in addition to the exposures to temporary contract and redundancy plan, thus this number ranged from 0 to 10.

2.3. Statistical analyses

All statistical analyses were performed using R statistical software. Workers were followed up from the date of interview until the first date of prescription of psychotropic medication, or until the end of follow-up (i.e. up to 2 years after survey wave), or until age 65, or until death, whichever came first. Mixed effects Cox regression models were used to examine the prospective associations between psychosocial work exposures at survey wave and the 2-year incident prescription of psychotropic medications, taking into account the non-independence of observations for workers surveyed twice. The proportional hazard assumption was checked by Schoenfeld residual plots and tests. Hazard ratios (HR) and 95 % confidence intervals (CI) were estimated before and after adjustment for the following covariates: gender, age (<30, [30–40], [40–50], ≥50 years), marital status (living with a partner or not), employee/self-employed worker, public/private sector, occupation (managers/professionals, technicians/associate professionals, clerks/service workers, blue collar workers), full/part-time work, and survey wave. Effect modifications by gender were tested by adding interaction terms between gender and psychosocial work exposures in Cox regression models. Analyses also explored the prospective associations of psychosocial work exposures with the 2-year incident prescription of each therapeutic class of psychotropic medications. Sensitivity analyses were performed with additional adjustment for (1) other occupational exposures (i.e. night work, biomechanical exposure, exposure to dangerous or toxic products), (2) chronic disease, (3) health behaviours (smoking, alcohol consumption, and body mass index), and (4) after exclusion of workers with a disabled/handicapped person in the household.

3. Results

A description of covariates and psychosocial work exposures among the 7695 workers of the study sample (4151 men and 3544 women) is presented in Tables S1-S2. There were gender differences in almost all covariates. There were also gender differences in the prevalence of most psychosocial work exposures.

The mean follow-up time was 1.8 years from the date of interview and there were 1541 incident cases of psychotropic medication prescription. The 2-year incidence of psychotropic medication prescription was higher in women (23.3 %) than in men (13.4 %). It increased with age and was higher among employees, technicians/associate professionals, clerks/service workers, and part-time workers (Table S3). All psychosocial work exposures, except low possibilities for learning new things, low satisfaction with salary and temporary contract, were associated with the 2-year incident prescription of psychotropic medications after adjustment for covariates (Table 1). A linear trend was found between the number of psychosocial work exposures and incident psychotropic medication prescription (Table 1). No gender interaction was found for these associations. Psychosocial work exposures were mainly associated with antidepressants and anxiolytics, except high quantitative demands, high tensions with the public and low freedom at work which were associated with incident prescription of each psychotropic therapeutic class (Table 2). Night work, biomechanical exposure, chronic disease, daily smoking, no alcohol consumption and disabled/handicapped person in the household were associated with incident prescription of psychotropic medications (Table S3). The results were unchanged in the sensitivity analyses (Table S4).

4. Discussion

4.1. Summary of study results

The results of this 2-year prospective study showed that various psychosocial work exposures increased the risk of subsequent psychotropic medication prescription and there was a dose-response association between the number of psychosocial work exposures and the risk of this outcome. Prospective associations were observed mainly for antidepressants and anxiolytics. No gender differences were found in these associations.

4.2. Comparison with the literature

Our results are in line with the conclusion of a systematic review and meta-analysis of prospective studies showing an association between high job demands and psychotropic medications (Milner et al., 2019).

The association between tensions with the public and psychotropic medications was studied in a previous French study based on self-reported psychotropic medications and no association was found (Lassalle et al., 2015). However, this association was in agreement with a Finnish prospective study showing an association between work-

Table 1
Psychosocial work exposures and 2-year incident prescription of psychotropic medications.

	Unadjusted models				Covariate-adjusted models[a]			
	N	Incident cases, n	HR (95 % CI)	p-Value	N	Incident cases, n	HR (95 % CI)	p-Value
Each psychosocial work exposure studied separately								
High quantitative demands	8611	1520	1.20 (1.08–1.33)	**<0.001**	8577	1516	1.24 (1.12–1.38)	**<0.001**
High tensions with the public	8536	1506	1.27 (1.13–1.42)	**<0.001**	8502	1502	1.23 (1.09–1.39)	**<0.001**
Low freedom at work	8561	1507	1.43 (1.25–1.63)	**<0.001**	8527	1503	1.37 (1.19–1.57)	**<0.001**
Low possibilities for learning new things	8610	1521	1.16 (1.05–1.29)	**0.005**	8576	1517	1.10 (0.99–1.24)	0.082
Low support from colleagues	8577	1515	1.24 (1.12–1.38)	**<0.001**	8543	1511	1.23 (1.10–1.37)	**<0.001**
Low recognition at work	8563	1514	1.21 (1.09–1.35)	**<0.001**	8529	1510	1.16 (1.04–1.29)	**0.007**
Low satisfaction with salary	8491	1500	1.15 (1.03–1.29)	**0.014**	8457	1496	1.10 (0.98–1.24)	0.096
High job insecurity	8570	1514	1.29 (1.10–1.53)	**0.002**	8537	1510	1.29 (1.09–1.53)	**0.003**
Temporary contract	8681	1538	0.90 (0.76–1.06)	0.199	8649	1535	0.88 (0.73–1.05)	0.154
Redundancy plan within the last 12 months	8522	1511	1.18 (1.00–1.40)	0.052	8490	1507	1.20 (1.01–1.43)	**0.038**
Number of psychosocial work exposures	8479	1492		**<0.001**[b]	8445	1488		**<0.001**[b]
0	437	58	1 (Ref.)		436	58	1 (Ref.)	
1	1072	153	1.07 (0.78–1.46)	0.680	1070	152	1.03 (0.75–1.40)	0.876
2	1580	247	1.18 (0.87–1.58)	0.285	1574	247	1.14 (0.85–1.54)	0.386
3	1761	291	1.26 (0.94–1.69)	0.118	1752	291	1.22 (0.91–1.63)	0.194
4	1636	298	1.40 (1.05–1.87)	**0.024**	1628	297	1.32 (0.99–1.78)	0.062
5	1085	226	1.64 (1.22–2.22)	**0.001**	1079	226	1.58 (1.17–2.14)	**0.003**
≥6	908	219	1.96 (1.45–2.65)	**<0.001**	906	217	1.79 (1.32–2.43)	**<0.001**

HR: hazard ratio, CI: confidence interval.
Bold: significant at $p<0.05$.
[a] Adjustment for the following covariates: gender, age, marital status, employee/self-employed worker, private/public sector, occupation, full/part-time work, and survey wave.
[b] Test for linear trend in the HR for psychotropic medication prescription with the number of psychosocial work exposures.

Table 2
Psychosocial work exposures and 2-year incident prescription of each therapeutic class of psychotropic medications after adjustment for covariates.

	N	Antidepressants (ATC code: N06A)			Anxiolytics (ATC code: N05B)			Hypnotics/Sedatives (ATC code: N05C)		
		Incident cases, n	Covariate-adjusted HR (95 % CI)[a]	p-Value	Incident cases, n	Covariate-adjusted HR (95 % CI)[a]	p-Value	Incident cases, n	Covariate-adjusted HR (95 % CI)[a]	p-Value
Each psychosocial work exposure studied separately										
High quantitative demands	8577	444	1.35 (1.12–1.64)	**0.002**	1205	1.20 (1.07–1.35)	**0.003**	383	1.28 (1.04–1.58)	**0.020**
High tensions with the public	8502	440	1.29 (1.04–1.60)	**0.020**	1194	1.20 (1.05–1.37)	**0.008**	379	1.39 (1.11–1.75)	**0.005**
Low freedom at work	8527	440	1.67 (1.33–2.09)	**<0.001**	1193	1.30 (1.12–1.51)	**<0.001**	383	1.30 (1.00–1.69)	**0.050**
Low possibilities for learning new things	8576	445	1.25 (1.02–1.52)	**0.033**	1205	1.13 (1.00–1.28)	0.060	385	0.79 (0.63–0.98)	**0.030**
Low support from colleagues	8543	442	1.25 (1.03–1.52)	**0.024**	1202	1.27 (1.12–1.43)	**<0.001**	382	1.11 (0.90–1.37)	0.341
Low recognition at work	8529	442	1.45 (1.19–1.76)	**<0.001**	1201	1.15 (1.02–1.30)	**0.019**	383	1.10 (0.90–1.36)	0.353
Low satisfaction with salary	8457	438	1.37 (1.11–1.70)	**0.004**	1190	1.13 (1.00–1.29)	0.059	380	1.00 (0.80–1.24)	0.969
High job insecurity	8537	443	1.46 (1.09–1.95)	**0.011**	1201	1.35 (1.12–1.62)	**0.002**	384	1.10 (0.78–1.55)	0.570
Temporary contract	8649	450	0.88 (0.63–1.22)	0.432	1219	0.86 (0.71–1.05)	0.146	388	1.02 (0.72–1.44)	0.913
Redundancy plan within last 12 months	8490	441	1.29 (0.95–1.75)	0.104	1200	1.25 (1.03–1.51)	**0.022**	383	1.13 (0.80–1.58)	0.490
Number of psychosocial work exposures	8445	437		**<0.001**[b]	1184		**<0.001**[b]	378		0.111[b]
0	436	12	1 (Ref.)		43	1 (Ref.)		16	1 (Ref.)	
1	1070	29	0.97 (0.49–1.92)	0.931	118	1.09 (0.76–1.56)	0.630	46	1.12 (0.63–2.00)	0.698
2	1574	71	1.62 (0.87–3.00)	0.129	199	1.26 (0.90–1.77)	0.183	63	1.05 (0.60–1.83)	0.871
3	1752	88	1.79 (0.97–3.30)	0.063	231	1.29 (0.92–1.81)	0.133	76	1.17 (0.67–2.02)	0.588
4	1628	87	1.87 (1.01–3.45)	**0.047**	235	1.41 (1.01–1.97)	**0.045**	66	1.05 (0.60–1.84)	0.871
5	1079	67	2.21 (1.18–4.14)	**0.013**	184	1.70 (1.21–2.40)	**0.002**	65	1.66 (0.94–2.92)	0.078
≥6	906	83	3.26 (1.75–6.07)	**<0.001**	174	1.89 (1.33–2.67)	**<0.001**	46	1.32 (0.73–2.38)	0.354

HR: hazard ratio, CI: confidence interval.
Bold: significant at $p<0.05$.
[a] Adjustment for the following covariates: gender, age, marital status, employee/self-employed worker, private/public sector, occupation, full/part-time work, and survey wave.
[b] Test for linear trend in the HR for psychotropic medication prescription with the number of psychosocial work exposures.

related violence and antidepressants (Madsen et al., 2011).

The increased risk of psychotropic medications associated with job insecurity was consistent with the results of two cohort studies showing associations of job insecurity with any psychotropic drug use (Blomqvist

et al., 2020), antidepressants (Blomqvist et al., 2020; Rugulies et al., 2010) and anxiolytics (Blomqvist et al., 2020). The non-significant association between temporary employment and psychotropic medications in our study is consistent with the findings from the Helsinki Health cohort study focusing on antidepressants and sedatives (Laaksonen et al., 2012). In agreement with the association observed for redundancy plan, organisational changes especially employee lay-off (Jensen et al., 2019) and organisational downsizing (Kaspersen et al., 2017) were previously found to increase the risk of prescription of psychotropic drugs.

In contrast to our study showing an association between low freedom at work and all types of psychotropic medications, close measures related to low influence at work were associated neither with antidepressants among Danish and Swedish employees (Madsen et al., 2011; Magnusson Hanson et al., 2013), nor with anxiolytics and hypnotics (Madsen et al., 2011). Low possibilities for learning new things were found to be associated with antidepressants, while previous prospective studies observed no association for close measures, i.e. learning possibilities (Magnusson Hanson et al., 2013) and possibilities for development (Thielen et al., 2011). Low social support from colleagues increased the risk of prescription of antidepressants and anxiolytics and there was no gender-related difference in these associations, while no association was reported with anxiolytics among Danish employees (Madsen et al., 2011) and antidepressant use increased with low social support from colleagues among men only (Thielen et al., 2011). Job dissatisfaction was previously found to increase the risk of any psychotropic drugs, particularly of antidepressants among both genders and of sleeping pills among women (Laaksonen et al., 2012). In our study, low satisfaction with salary was associated with antidepressant medications.

Our study provided information about the impact of multiple psychosocial work exposures, that has received limited attention (Boot et al., 2024; Niedhammer et al., 2021). Only combined exposures related to the two work stress models of job strain or effort-reward imbalance have previously been reported in association with psychotropic medications (Milner et al., 2019).

Finally, among the studies that statistically tested gender-related interactions in the prospective associations between psychosocial work exposures and psychotropic medications, most reported no significant differences between genders (Blomqvist et al., 2020; Conway et al., 2025; Kouvonen et al., 2017; Lallukka et al., 2013; Lassalle et al., 2015; Madsen et al., 2011), which is consistent with our findings. One prospective study found an effect modification by gender for one of the six studied psychosocial working conditions, namely the association between poor workplace climate and antidepressants which was stronger among women (Laaksonen et al., 2012).

4.3. Strengths and limitations

The strengths of this study included: prospective design, study sample from a national population-based survey, high rates of response to the questionnaire on working conditions and of follow-up through linkage with the national database of healthcare reimbursements, study of the different therapeutic classes of psychotropic medications, use of data from two independent sources for exposures and outcome reducing reporting bias, statistical test of gender modification effects in the associations, and study of multiple exposures.

There were a number of limitations. Although the response and linkage rates were high, response and linkage biases remained possible. Psychosocial work exposures were not assessed by a validated questionnaire, and formal definition was missing for some items, which might lead to misclassification. The assessment of psychosocial work exposures and covariates was based on a single assessment at survey wave (baseline), however changes in these variables during follow-up were not available. The outcome was based on the national database of healthcare reimbursements, that includes all medications dispensed in pharmacies, but there was information neither on the medical reason for the prescription nor on compliance with psychotropic medications. Moreover, differential misclassification of the outcome might impact the results (Thielen et al., 2009). Based on available data, we excluded cases within the 6 months preceding survey wave, which might be considered as a too short period. Several covariates were taken into account in the main and sensitivity analyses, however a potential residual confounding bias could not be excluded. Our study might be underpowered to study gender-related interactions and the three therapeutic classes separately.

5. Conclusions

The results of this study may support that poor psychosocial working conditions increase the risk of subsequent psychotropic medication prescription, and show the importance of considering various aspects of the psychosocial work environment and multiple exposures. Further studies are needed to confirm these associations.

CRediT authorship contribution statement

Sandrine Bertrais: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Maël Quatrevaux:** Writing – review & editing, Software, Data curation. **Isabelle Niedhammer:** Writing – review & editing, Supervision, Project administration, Methodology, Funding acquisition, Conceptualization.

Ethics approval

The study was approved by French ethics committees: CNIL no 2010-003, 2012-055 and 2013-323, CESREES no 13087724 & CNIL no 2023-186.

Role of the funding source

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Declaration of competing interest

None.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jad.2025.121078.

Data availability

The used data are not available due to French ethical and legal restrictions.

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